

A decorative graphic on the left side of the slide consisting of two overlapping parallelograms. The front one is blue and the back one is light green. They are positioned diagonally, with the blue one partially covering the green one.

Training Deep Nets

Sadeep Jayasumana



Improving Accuracy

1. More training data
2. Better initialization, use pre-training/transfer learning
3. Better network architectures
4. Data augmentation
5. Prevent overfitting: weight decay, dropout, dropconnect
6. Batch normalization
7. Better optimisation algorithm



... more training data

<https://youtu.be/Ucp0TTmvqOE?t=7527>

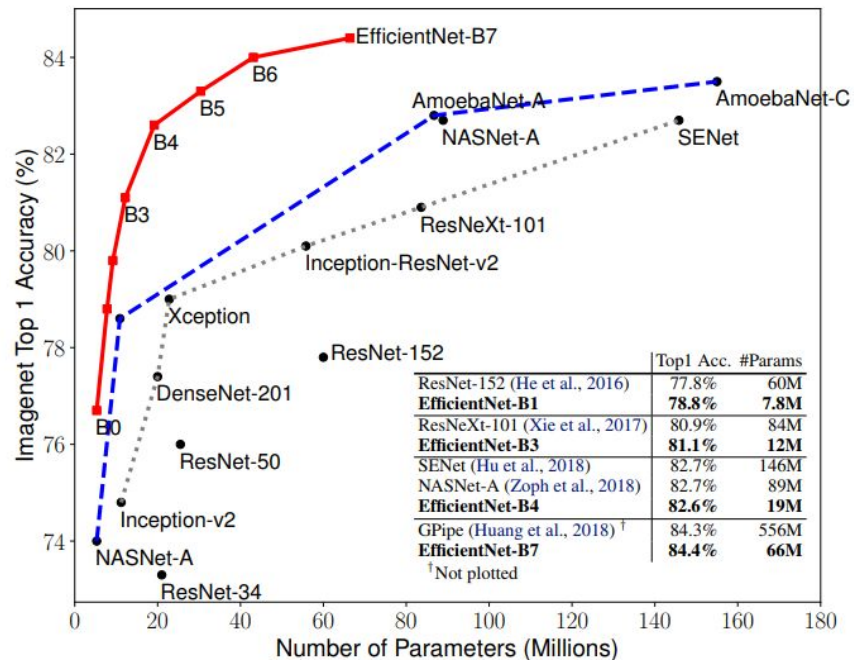


... better initialization

- Xavier initialization: <http://proceedings.mlr.press/v9/glorot10a/glorot10a.pdf>
- Use a pre-trained model
 - ImageNet pretraining
 - Unsupervised pretraining

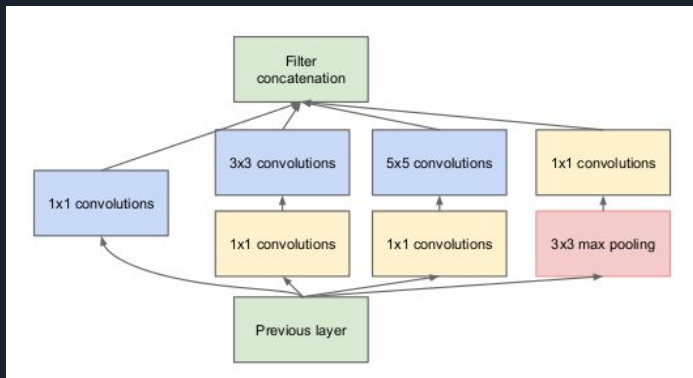
$$\text{Var}(w) = 2/(n_{in} + n_{out})$$

... better network architectures

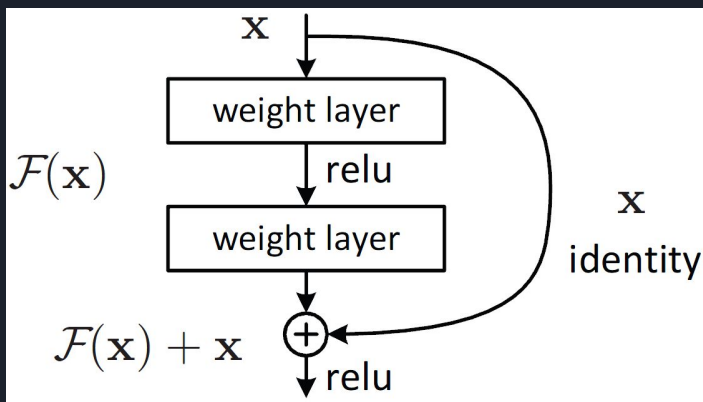


- Inception, Resnet architectures are still a fair bet
- Mobilenet architectures if you're using mobile phones
- Architecture can be learned from data as well!

... better network architectures

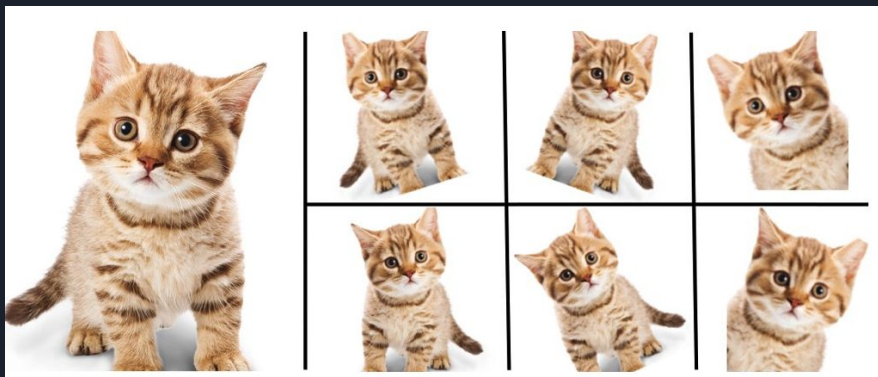


Inception networks



Resnet style networks/residual blocks

... data augmentation



- Rotating
- Flipping
- Cropping
- Adding noise
- Changing the colors slightly



... prevent overfitting

- Weight decay
- Dropout
- Dropconnect
- Early-stopping

... batch normalization

Input: Values of x over a mini-batch: $\mathcal{B} = \{x_{1\dots m}\}$;

Parameters to be learned: γ, β

Output: $\{y_i = \text{BN}_{\gamma, \beta}(x_i)\}$

$$\mu_{\mathcal{B}} \leftarrow \frac{1}{m} \sum_{i=1}^m x_i \quad // \text{ mini-batch mean}$$

$$\sigma_{\mathcal{B}}^2 \leftarrow \frac{1}{m} \sum_{i=1}^m (x_i - \mu_{\mathcal{B}})^2 \quad // \text{ mini-batch variance}$$

$$\hat{x}_i \leftarrow \frac{x_i - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}} \quad // \text{ normalize}$$

$$y_i \leftarrow \gamma \hat{x}_i + \beta \equiv \text{BN}_{\gamma, \beta}(x_i) \quad // \text{ scale and shift}$$



... better optimization algorithms

- Stochastic gradient descent (SGD)
- SGD with momentum
- RMSProp
- Adagrad
- ADAM